hw2_submit

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1

Eigval for AA^T: $AA^T = \begin{bmatrix}1&1\\1&1\end{bmatrix}\begin{bmatrix}1&1\\1&1\end{bmatrix} = \begin{bmatrix}2&2\\2&2\end{bmatrix}$ $\quad \begin{vmatrix}2-\lambda & 2\\ 2 & 2-\lambda\end{vmatrix} = (2-\lambda)(2-\lambda)-4 = 4-4\lambda+\lambda^2-4 = \lambda(\lambda-4)$ $\quad \lambda_1 = 4$, $\lambda_2 = 0$

For V, get eigvec of A^TA: $A^TA = AA^T$ b/c $A^T = A$ so V_1: $\begin{bmatrix}-2&2\\2&2\end{bmatrix}$ $\begin{bmatrix}1&-1\\0&0\end{bmatrix}$ $v_1 = \begin{pmatrix}1\\1\end{pmatrix}$ normalize to get $\begin{pmatrix}\frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}}\end{pmatrix}$

V_2: $\begin{bmatrix}2&2\\2&2\end{bmatrix} \rightarrow \begin{pmatrix}1&1\\0&0\end{pmatrix}$ $v_2 = \begin{pmatrix}1\\-1\end{pmatrix}$ norm $\frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}$

So $V = \boxed{\begin{bmatrix}\frac{1}{\sqrt{2}}&\frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}}&\frac{-1}{\sqrt{2}}\end{bmatrix} = V^*}$

U = eigvecs of AA^T: $AA^T = A^TA$ so eigvecs are same

$U = \begin{bmatrix}\frac{1}{\sqrt{2}}&\frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}}&-\frac{1}{\sqrt{2}}\end{bmatrix}$ $\qquad \Sigma = \begin{bmatrix}\sqrt{\lambda_1}&0\\0&\sqrt{\lambda_2}\end{bmatrix} = \begin{bmatrix}2&0\\0&0\end{bmatrix}$

2

a)

Given: A =QBQ^T where QQ^T=Q^TQ=I

Let λ be a singular value. Then $\exists x \neq 0$ st $A^T Ax = \lambda x$

$\implies (QBQ^T)^T(QBQ^T)x = \lambda x$

$\implies (QB^TQ^T)(QBQ^T)x = \lambda x$

$\implies QB^TBQ^Tx = \lambda x$

$\implies B^TBQ^Tx = Q^T\lambda x$

$\implies B^TBQ^Tx = \lambda Q^Tx$

Since Q is orthogonal, we know that Q and Q^T are nonzero and since x is nonzero, we know that $Q^Tx \neq 0$ so let $y = Q^Tx$ where $y \neq 0$

So $B^TBy = \lambda y$ therefore, λ is a singular value for B

b)

Converse: If they have the same singular values, then they are orthogonally similar

Let A=[4] and B=[-4]. The have the same singular values, 4. For A and B to be orthogonally similar, we need Q to be positive and Q^T to be negative (or vice versa). However if Q is positive then its transpose is also positive and if Q is negative then its transpose is also negative. So it is not possible for A and B to be orthogonally similar

3

In SVD, the choice of the singular values being positive is conventional

A is nonsingular so we know that it is invertable and has rank m
Let λ be an eigen value for A. Since A is nonsingular, $\lambda \neq 0$
This means that $Ax = \lambda x$ so multiplying by A^-1 to both sides,
$x = A^{-1}\lambda x \implies A^{-1}x = \frac{1}{\lambda}x$ so we know that the eigenvalues of A^{-1} are the reciprocal of the eigenvalues of A.
Since A is nonsingular, we know that $\sigma_1 \geq \ldots \geq \sigma_m > 0$.
By what we showed previously, we know that the singular values of A^{-1} are the reciprocal of the singular values of A

> ie $\frac{1}{\sigma_1} \ldots \frac{1}{\sigma_m}$

Ssince σ_m was the smallest, $\frac{1}{\sigma_m}$ is the largest singular value for A^-1 therefore
$||A^{-1}||_2 = \frac{1}{\sigma_m}$

A similar proof is that $A^{-1} = (U\Sigma V^*)^{-1} = V\Sigma^{-1}U^{-1}$ and since Σ is a diagonal matrix, its inverse will be the reciprocal of its values. And the largest singular value of Σ^{-1} will be the reciprocal of the smallest singular value of Σ (σ_m) so $||A^{-1}||_2 = \frac{1}{\sigma_m}$

4

```
A=[1 2 3 4; 5 6 7 8; 9 10 11 12; 1 1 1 1; 3 2 1 0]
[U, S, V] = svd(A, 'econ');

rank = sum(diag(S) > 1e-5)
A_2_norm = max(S, [], 'all')

fprintf("The orthonormal basis of the range is the column vectors of the
following matrix")
range_A= U(:, 1:rank)

fprintf("The orthonormal basis of the null space is the column vectors of
the following matrix")
null_A = V(:, rank+1:end)
```

```
>> q4

A =

     1     2     3     4
     5     6     7     8
     9    10    11    12
     1     1     1     1
     3     2     1     0


rank =

     2


A_2_norm =

   25.6545

The orthonormal basis of the range is the column vectors of the following matrix
range_A =

   -0.2043   -0.5076
   -0.5137   -0.1885
   -0.8230    0.1306
   -0.0773    0.0798
   -0.1051    0.8267

The orthonormal basis of the null space is the column vectors of the following matrix
null_A =

   -0.3590    0.4137
    0.7870   -0.2840
   -0.4970   -0.6731
    0.0690    0.5434
```

5

For this, assume $Q = \hat{Q}$ and $R = \hat{R}$. I got lazy

$\Longrightarrow$

Let rankA=n, then rankA^T = n so A^TA is nonsingular.

Then we know that $det(A^TA) \neq 0$

We know that $det(A^TA) = det((QR)^TQR) = det(R^TQ^TQR)$. Since Q is made up of orthonormal columns, $Q^TQ = I_m$ so $= det(R^TR) = det(R^T)det(R) \neq 0$

This is only possible if and only if $det(R^T) \neq 0$ and $det(R) \neq 0$

So we now have $det(R) \neq 0$, since R is a triangular matrix, we know that the determinant of R is the product of its main diagonal and since determinant of R is not 0, there must not be a 0 along the main diagonal. Proof complete

$\Leftarrow$

R has all nonzero entries on the main diagonal

> This implies that R is nonsingular so rank(R)=n

By rank inequalities: rank(A) = Rank(QR) >= rank(Q) + rank(R) - n

Since Q is an orthogonal matrix, rankQ= min{m,n} = n and rank(R)=n so rank(A) >= n + n - n = n

So now we have rank(A) >= n. Since A is m x n matrix, we know its max rank is n thus rank(A)=n

6

For this, assume $Q = \hat{Q}$ and $R = \hat{R}$. I got lazy

Show that the columns of Q form an orthonormal basis for the Range ofA.

So we know that the cols of Q are orthonormal by the defn of QR factorization and we know the cols of Q form the range(Q) by base defn. WTS range(Q)=range(A). Will do this by containment and reverse containment

$range(Q) \supseteq range(A)$
Let $x \in Range(A)$ where $x \neq 0$, since rankA=n $\exists y \neq 0$ st Ay=x $\implies QRy = x$, since $y \neq 0, Ry = z \neq 0$ so $\forall x \in Range(A) \exists z \ st \ Qz = x$ so $x \in range(Q)$ so $range(A) \subseteq range(Q)$

$range(Q) \subseteq range(A)$
Let $x \in Range(Q)$ where $x \neq 0$, then $\exists y \neq 0$ st Qy=x. Since R is upper triangular with nonzero entries along the main diagonal, it is nonsingular and therefore invertable. So we can write $Q = AR^{-1}$ so $Qy = AR^{-1}y = x$ so if we let $z = R^{-1}y, Az = x$ so then $\forall x \in Range(Q), \exists z \ st \ Az = x$ so $range(Q) \subseteq range(A)$

So $range(Q) \subseteq range(A) \subseteq range(Q)$
So range(Q) = range(A)

7

Code:

```
function [Q, R] = clgs(A)

    n = size(A, 2);

    R = zeros(n);
    Q = zeros(size(A,1), n);
    for j = 1:n
        v = A(:, j);


        for i = 1:(j-1)
            R(i,j) = Q(:, i)' * A( :, j);
            v = v - R(i, j) * Q(:, i);
```

```
        end

        R(j,j) = norm(v);
        Q(:, j) = v / R(j,j);

    end
end

function [Q, R] = mgs(A)

    n = size(A, 2);

    R = zeros(n);
    Q = zeros(size(A,1), n);

    V = zeros(size(A));

    for i = 1:n
        V(:, i) = A(:, i);
    end


    for i = 1:n

        R(i , i) = norm(V(:, i));
        Q( :, i) = V(:, i) / R(i,i);

        for j = (i+1):n
            R(i , j) = Q(:,i)' * V(:, j);
            V( :, j) = V( :, j) - R(i, j) * Q(:, i);
        end
    end
end

epsillons = [1e-4, 1e-6, 1e-8];
format long

for e = epsillons
    A = [1, 1, 1; e, 0, 0; 0, e, 0; 0, 0, e];

    fprintf("Q and R for epsilon = %f using classical GSP", e)
    [Q, R] = clgs(A)
    fprintf("QR:")
    Q*R
    fprintf("Q^TQ:")
    Q' * Q
```

```
    fprintf("Q and R for epsilon = %f using modified GSP", e)
    [Q, R] = mgs(A)
    fprintf("QR:")
    Q * R
    fprintf("Q^TQ:")
    Q' * Q
end
```

```
…/782/hw2 main* ❯ matlab -batch q7
Q and R for epsilon = 0.000100 using classical GSP
Q =

   0.999999995000000   0.000070710677866   0.000040824828876
   0.000099999999500  -0.707106775883247  -0.408248287161463
                   0   0.707106782954315  -0.408248290364194
                   0                   0   0.816496581608140


R =

   1.000000005000000   0.999999995000000   0.999999995000000
                   0   0.000141421355884   0.000070710677866
                   0                   0   0.000122474487037

QR:
ans =

   1.000000000000000   1.000000000000000   1.000000000000000
   0.000100000000000  -0.000000000000000                   0
                   0   0.000100000000000                   0
                   0                   0   0.000100000000000

Q^TQ:
ans =

   1.000000000000000   0.000000000000277   0.000000000000160
   0.000000000000277   1.000000000000000  -0.000000002264673
   0.000000000000160  -0.000000002264673   1.000000000000000

Q and R for epsilon = 0.000100 using modified GSP
Q =

   0.999999995000000   0.000070710677866   0.000040824829036
   0.000099999999500  -0.707106775883247  -0.408248288762828
                   0   0.707106782954315  -0.408248288762829
                   0                   0   0.816496581608140


R =

   1.000000005000000   0.999999995000000   0.999999995000000
                   0   0.000141421355884   0.000070710677588
                   0                   0   0.000122474487037
```

```
QR:
ans =

   1.000000000000000   1.000000000000000   1.000000000000000
   0.000100000000000  -0.000000000000000                   0
                   0   0.000100000000000                   0
                   0                   0   0.000100000000000

Q^TQ:
ans =

   1.000000000000000   0.000000000000277   0.000000000000160
   0.000000000000277   1.000000000000000  -0.000000000000000
   0.000000000000160  -0.000000000000000   1.000000000000000

Q and R for epsilon = 0.000001 using classical GSP
Q =

   0.999999999999500   0.000000707169643   0.000000408248287
   0.000001000000000  -0.707106781186017  -0.408211996414818
                   0   0.707106781186724  -0.408284583437063
                   0                   0   0.816496579852289


R =

   1.000000000000500   0.999999999999500   0.999999999999500
                   0   0.000001414213562   0.000000707169643
                   0                   0   0.000001224744873

QR:
ans =

   1.000000000000000   1.000000000000000   1.000000000000000
   0.000001000000000                   0                   0
                   0   0.000001000000000                   0
                   0                   0   0.000001000000000

Q^TQ:
ans =

   1.000000000000000   0.000000000062862   0.000000000036290
   0.000000000062862   1.000000000000000  -0.000051326775655
   0.000000000036290  -0.000051326775655   1.000000000000000
```

```
Q and R for epsilon = 0.000001 using modified GSP
Q =

   0.999999999999500   0.000000707169643   0.000000408284584
   0.000001000000000  -0.707106781186017  -0.408248290463693
                   0   0.707106781186724  -0.408248290463693
                   0                   0   0.816496580927794


R =

   1.000000000000500   0.999999999999500   0.999999999999500
                   0   0.000001414213562   0.000000707106781
                   0                   0   0.000001224744871

QR:
ans =

   1.000000000000000   1.000000000000000   1.000000000000000
   0.000001000000000                   0                   0
                   0   0.000001000000000                   0
                   0                   0   0.000001000000000

Q^TQ:
ans =

   1.000000000000000   0.000000000062862   0.000000000036294
   0.000000000062862   1.000000000000000  -0.000000000000000
   0.000000000036294  -0.000000000000000   1.000000000000000

Q and R for epsilon = 0.000000 using classical GSP
Q =

   1.000000000000000                   0                   0
   0.000000010000000  -0.707106781186548  -0.707106781186548
                   0   0.707106781186548                   0
                   0                   0   0.707106781186548


R =

   1.000000000000000   1.000000000000000   1.000000000000000
                   0   0.000000014142136                   0
                   0                   0   0.000000014142136
```

```
QR:
ans =

   1.000000000000000   1.000000000000000   1.000000000000000
   0.000000010000000                   0                   0
                   0   0.000000010000000                   0
                   0                   0   0.000000010000000

Q^TQ:
ans =

   1.000000000000000  -0.000000007071068  -0.000000007071068
  -0.000000007071068   1.000000000000000   0.500000000000000
  -0.000000007071068   0.500000000000000   1.000000000000000

Q and R for epsilon = 0.000000 using modified GSP
Q =

   1.000000000000000                   0                   0
   0.000000010000000  -0.707106781186548  -0.408248290463863
                   0   0.707106781186548  -0.408248290463863
                   0                   0   0.816496580927726


R =

   1.000000000000000   1.000000000000000   1.000000000000000
                   0   0.000000014142136   0.000000007071068
                   0                   0   0.000000012247449

QR:
ans =

   1.000000000000000   1.000000000000000   1.000000000000000
   0.000000010000000                   0                   0
                   0   0.000000010000000                   0
                   0                   0   0.000000010000000

Q^TQ:
ans =

   1.000000000000000  -0.000000007071068  -0.000000004082483
  -0.000000007071068   1.000000000000000  -0.000000000000000
  -0.000000004082483  -0.000000000000000   1.000000000000000

…/782/hw2 main* ❯ 
```

The modified GSP was significantly better. This is evident at the smaller epsilons where Q^TQ is much close to I_n for the modified GSP compared to the regular